Operations Research for Health Care

Lesson 7/15

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Outline of this lecture

Hierarchical health care location problems

Analytical and improved 3-hierarchical system

Hierarchical with referral of patients

Intermediate facility location

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Review of hierarchical health care location [Ortiz-Astorquiza et al., 2018]

Multiple layer configuration: Network with k-levels: nodes represent demand site and facilities. Patients on level 0, primary-care on leve 1, ... highest level on level k.
Classification:

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- **Flow pattern**: single-flow, multi-flow, bidirectional-flow. (referral system).
- **Service**: successively inclusive and exclusive (nested and non-nested).
- **Spatial configuration**: Coherent or non-coherent system.
- **Objectives**: minimizing total transportation costs/distances (*median*), cost for opening (*fixed charge*), maximizing coverage (*maximal covering*), other.

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Given a set of patients that demand one or k services and a set of facilities associated with k services, select a set of facilities to open so that each patient receives the requested service(s), while optimizing a criteria.

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Analytical 3-hierarchical system [Calvo and Marks, 1973]

Motivation:

- Locational criteria: hierarchy of social, economic and political attributes.
- Hospitals, health centers, nursing homes, and private clinics that are planned, organized, and operated independently of each other. The result has been a large duplication of services, inefficient utilization of resources, and imbalance in facility locations.
- The critical issue: *where should the facilities be located in a region in order to satisfy the health care requirements of its population and comply with the social, economic, and political criteria relevant to the region*?

Analytical 3-hierarchical system [Calvo and Marks, 1973]

The problem:

- **Regional hospital** (specialized treatment), **district hospitals** (intermediate care, *lacking the specialized treatment of regional facility*), **rural hospitals** (outpatient care and general inpatient care). There is intensive communication and *patient referral* between them. It is a successively inclusive property.

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- Within the **hospital**, the physicians' offices are to be located, with the necessary paramedical services and the appropriate communication devices and technology sources. The **community hospitals** offer essentially the same services, but of lesser degree, not offering the specialized services.

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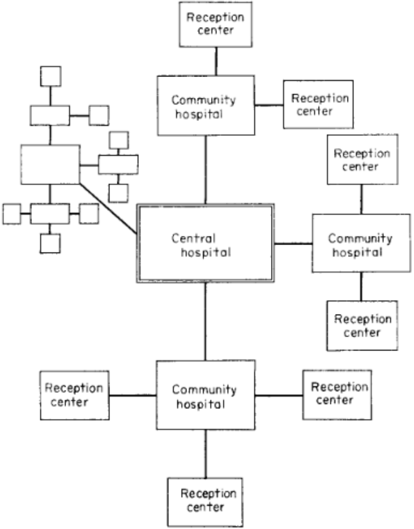
- The **reception centers** will provide the "*entry points*" to the entire system for the population of the region. Principally, they will be staffed with *paramedical personnel and public health nurses providing auxiliary treatment, checkups and screening tests, and with social workers providing assistance with the community's social problems*. These centers will provide the public with greater accessibility to health care services.

Analytical 3-hierarchical system [Calvo and Marks, 1973]



Analytical 3-hierarchical system [Calvo and Marks, 1973]

Measure of effectiveness. Geografical, financial accessibility, and effectiveness in meeting the demand and needs of the region. Three sectors of society are involved:

- **The user**: patient population (*time and cost to travel, service cost directly or indirectly, through taxes or insurances*);

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- **The community**: resident population and local businesses, socio-economic impacts. *Traffic, environment, political considerations and platforms of debate*.

Analytical 3-hierarchical system [Calvo and Marks, 1973]

Optimization Parameters: The **region** is disaggregated into n *planning areas*, each composed of s *population groups*. E.g., population group k in community $i(i = 1, 2, ..., n)$ will demand order of service $k(k = 1, ..., s)$. The allocation of demand should be conducted so as to satisfy the health care needs of the region and optimize the relevant planning criteria.

Let a_{ik} be the number of people in group k residing in area i and requiring order of services k. The total population in area i is

$$a_i = \sum_{k \in 1..s} a_{ik}, \quad \forall i \in 1..n$$

Analytical 3-hierarchical system [Calvo and Marks, 1973]

All population groups within an area have to be fully assigned to the corresponding facility-types.

$$\sum_{k \in 1}^{k*} \sum_{j \in 1}^{n} x_{ijk} = 1 \quad \forall \quad x_{ijk*} > 0 \tag{2}$$

$$\sum_{j \in 1..n} x_{ijk} = 1, \quad \forall \quad k* < k < s, k* \in 1..s, i \in 1.., n \tag{3}$$

Constraints (2) and (3) depend on *variable* K_i, obtained only after the facilities are located. A *chicken-and-egg* dilemma (impossible) to resolve [Tien et al., 1983].

a) (2) is replaced by (8), which assures that individuals requiring different types of services cannot obtain these services from the same facility, and b) the range in (9) is not dependent on a K_h as is the case in (3) [Tien et al., 1983].

$$\sum_{k \in 1..s} x_{ijk} \leq 1 \quad \forall \quad i \in 1..n, j \in 1..n \tag{8}$$

Analytical 3-hierarchical system [Calvo and Marks, 1973]

Assignment to only those areas which assign to themselves, for each order k facility. This avoids the possibility of a population group a_{ik} being assigned to area j, where there is no self assignment, that is no facility located there

$$x_{jjk} > x_{ikj}, \quad \forall \quad i = 1..n, j \in 1..n, k \in 1..s \tag{4}$$

A specific number of order k facilities (for $k = 1, 2, ..., s$) that may assign areas to themselves.

$$\sum_{i \in n} x_{iik} = m_k, \quad \forall \quad k = 1..s \tag{5}$$

$$x_{ijk} = \{0, 1\} \quad \forall \quad i \in 1..n, j \in 1..n, k = 1..s \tag{6}$$

As an alternative to a fixed number of facilities, the optimization may be carried out under a specified budget or investment level.

Analytical 3-hierarchical system [Calvo and Marks, 1973]

USER:

- minimize Weighted Distance: $\sum_{i\in 1..n}\sum_{j\in 1..n} d_{ij}\sum_{k\in 1..s} a_{ik}x_{ijk}$
- minimize Total Cost: $\sum_{i\in 1..n}\sum_{j\in 1..n} c_{ij}\sum_{k\in 1..s} a_{ik}x_{ijk}$

SERVICE:

- F_{jk}: (fixed cost for building facility k in area j)
- V_jk: (service and expansion cost for operating facility k in area j)
- S_jk: size of k facility, which depends on demand, i.e,
  $S_{jk} = \sum_{i\in 1..n} a_{ik}x_{ijk}, \quad \forall j, k$
- Facility Cost $= \sum_{j\in 1..n}\sum_{k\in 1..s} p_{jk} * F_{jk} * x_{ijk} + V_{jk}\sum_{i\in 1..n} a_{ik} * x_{ijk}$
- $l_k \leq \sum_{1..n} a_{ik} * x_{ijk} \leq u_k$ (lower and upper bound of capacity)

Objective Function:

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Objective Function:

$$minimize \sum_{i\in 1..n}\sum_{j\in 1..n} c_{ij}\sum_{k\in 1..s} a_{ik}x_{ijk} + \sum_{j\in 1..n}\sum_{k\in 1..s} p_{jk}F_{jk}x_{ijk} + V_{jk}\sum_{i\in 1..n} a_{ik}x_{ijk} \quad (1)$$

Analytical 3-hierarchical system [Calvo and Marks, 1973]

LOCATION PLANNING PROCESS: it is necessary to place these models in a general planning framework (see fig 4 for discussion).

1. Define location problem

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2. Define locational social, political and economical criteria
3. Optimize and conduct sensitivity analysis

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LOCATION PLANNING PROCESS: it is necessary to place these models in a general planning framework (see fig 4 for discussion).

1. Define location problem
2. Define locational social, political and economical criteria
3. Optimize and conduct sensitivity analysis
4. Evaluate "efficient solution". Satisfied? End, Else return to (2)

Analytical 3-hierarchical system [Calvo and Marks, 1973]

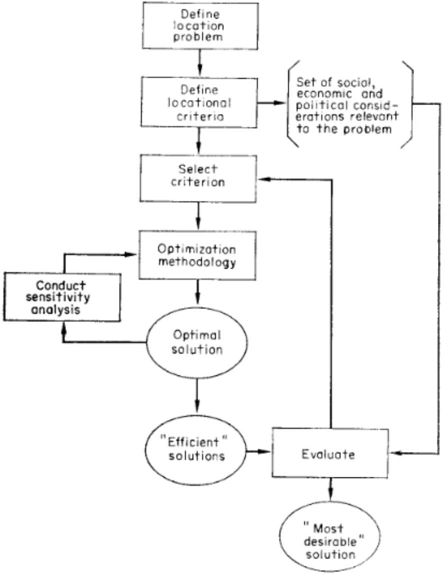

FIG. 4. Health facility location planning process.

Improved 3-hierarchical system [Tien et al., 1983]

$$\text{min.} \sum_{i \in 1..n} \sum_{j \in 1..n} d_{ij} \sum_{k \in 1..s} a_{ik} * x_{ijk} \tag{13}$$

$$\text{s.t.:} \sum_{j \in 1..n} x_{ijk} = 1, \quad \forall i, k \tag{14}$$

$$x_{jjk} \geq x_{ijk}, \quad \forall i, j, k \tag{15}$$

$$\sum_{i \in 1..n} x_{iik} = \sum_{r = k..s} m_r \quad \forall k \tag{16}$$

$$x_{ijk} \in \{0, 1\} \quad \forall i \in I, j \in J, k \in K$$

Outline of this lecture

Hierarchical with referral of patients [Narula and Ogbu, 1979]

- **Motivation**: A patient who is not completely cured at a health center may be referred to a hospital or a medical center for treatment; and a patient who first goes to a hospital may be referred to a medical center, if necessary. A medical center is equipped to treat all cases. This system as successfully implemented in Cuba.
- Optimal service radious (distance) for each type of facility: net social benefit per capita = cost per capita (Schultz, 1970), but not realistic, because population is not homogeneous, and referral was not considered.
- [Calvo and Marks, 1973] assumed that the patients at each population center can be divided into k(=3) groups according to their health care needs. They also did not allow for any referrals of the patients.
- **Offer**: In this paper, we allow the referral of $\theta(0 \leq \theta \leq 1)$ proportion of patients from a health center to a hospital.

Hierarchical [Ahmadi-Javid et al., 2017]

I : The set of demand points.
K : The set of candidate locations for a level-1 PCF (e.g., clinics).
J : The set of candidate locations for a level-2 PCF (e.g., hospitals).

- $d1_{ik}$: The travel time from demand point $i \in I$ to a level-1 PCF in a candidate location $k \in K$.
- $d2_{kj}$: The travel time from a level-1 PCF in a candidate location $k \in K$ and a level-2 PCF in a candidate location $j \in J$.

- w_i: The population size, demand point at $i \in I$.
- p: The number of a level-1 PCF locations to be established.
- q: The number of a level-2 PCF locations to be established.
- $C1_k$: The capacity of a level-1 PCF in candidate location $k \in K$.
- $C2_j$: The capacity of a level-2 PCF in candidate location $j \in J$
- θ_k: Proportion of level-1 PCF patients at candidate location $k \in K$ referred to a level-2 PCF.

Hierarchical [Ahmadi-Javid et al., 2017]

- $x1_k \in \{0, 1\}$: 1, if a level-1 PCF is established at candidate location $k \in K$; 0 otherwise.
- $x2_j \in \{0, 1\}$: 1, if a level-2 PCF is established at candidate location $j \in J$; 0 otherwise.
- $u_i k$: The flow of patients between demand point $i \in I$ and a level-1 PCF at candidate location $k \in K$
- $v_k j$: The flow of patients between demand point $k \in K$ and a level-2 PCF at candidate location $j \in J$

Hierarchical [Ahmadi-Javid et al., 2017]

$$\text{min.} \sum_{i \in I} \sum_{k \in K} d1_{ik} u_{ik} + \sum_{k \in K} \sum_{j \in J} d2_{kj} v_{kj} \tag{28}$$

$$\text{s.t.:} \sum_{k \in K} u_{ik} = w_i \quad \forall i \in I \tag{29}$$

$$\sum_{j \in J} v_{kj} = \theta_k \sum_{i \in I} u_{ik} \quad \forall k \in K \tag{30}$$

$$\sum_{i \in I} u_{ik} \leq C1_k x1_k \quad \forall k \in K \tag{31}$$

$$\sum_{k \in K} v_{kj} \leq C2_j x2_j \quad \forall j \in J \tag{32}$$

Hierarchical [Ahmadi-Javid et al., 2017]

$$\sum_{k \in K} x1_k = p \tag{33}$$

$$\sum_{j \in J} x2_j = q \tag{34}$$

$$x1_k \in \{0,1\} \quad \forall k \in K$$

$$x2_j \in \{0,1\} \quad \forall k \in K$$

$$u_{ik} \geq 0 \quad \forall i \in I, k \in K$$

$$v_{kj} \geq 0 \quad \forall k \in K, j \in J$$

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Intermediate facility location

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Scarce literature on health care.

- Publications are applied to warehouses, airport, and solid waste management
  - Kuehn, A. A.,& Hamburger, M. J. (1963). A heuristic program for locating warehouses. Management science, 9(4), 643-666.
  - Helms, B. P., & Clark, R. M. (1971). Locational models for solid waste management. Journal of the Urban Planning and Development Division, 97(1), 1-13.
  - Kanafani, A. (1972). Location models for off-airport terminals. Transportation Research, 6(4), 371-379.
  - Mirchandani, P. B., & Odoni, A. R. (1979). Locating new passenger facilities on a transportation network. Transportation Research Part B: Methodological, 13(2), 113-122.
- **We propose**: an adaptation of [Kuehn and Hamburger, 1963] to health care.
- Primary health care (PHC) units are fixed, number of hospitals are also fixed. There are candidate locations of intermediate health care units (IHC).
- Patients adopt a minimum-cost flow form PHC to hospital using best IHC units.

Intermediate facility location [Kuehn and Hamburger, 1963]

H : Types of Patients (cronic or acute).
I : Primary health care units (FIXED).
J : Intermediate facility *candidates* location.
K : Hospital units (FIXED).

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- A_{hij}: Travel cost/patients $h \in H$, from PHC $i \in I$ to IHC $j \in J$.
- B_{hjk}: Travel cost/patients $h \in H$, from IHC $j \in J$ to Hospital $k \in K$.
- S_{hj}: Variable cost of IHC $j \in J$, per patients $h \in H$.
- F_j: Fixed cost per period for operating IHC $j \in J$.
- Q_{hk}: Patients $h \in H$, requiring service at Hospital $h \in H$.
- W_j: Capacity of IHC j.
- Y_{hi}: Capacity of PHC i meet health care requirements of patients h.

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- x_{hijk}: Patients h, flow from PHC i to IHC j to Hospital k.
- z_j: If IHC j is used (1) or not (0).

Intermediate facility location [Kuehn and Hamburger, 1963]

$$\text{min.} \sum_{h}^{H}\sum_{i}^{I}\sum_{j}^{J}\sum_{k}^{K}(A_{hij} + B_{hjk})x_{hijk} + \sum_{j}^{J} F_j z_j + \sum_{h}^{H}\sum_{i}^{I}\sum_{j}^{J}\sum_{k}^{K} S_{hj}x_{hijk} \tag{1}$$

Intermediate facility location [Kuehn and Hamburger, 1963]

$$\text{min. } \sum_{h}^{H}\sum_{i}^{I}\sum_{j}^{J}\sum_{k}^{K}(A_{hij}+B_{hjk})x_{hijk}+\sum_{j}^{J}F_j z_j+\sum_{h}^{H}\sum_{i}^{I}\sum_{j}^{J}\sum_{k}^{K}S_{hj}x_{hijk} \tag{1}$$

$$\text{s.t.: } \sum_{i\in I}\sum_{j\in J}x_{hijk}=Q_{hk} \quad \forall h\in H, k\in K \tag{2}$$

$$\sum_{j\in J}\sum_{k\in K}x_{hijk}\le Y_{hi} \quad \forall h\in H, i\in I \tag{3}$$

$$\sum_{h\in H}\sum_{i\in I}\sum_{k\in K}x_{hijk}\le W_j z_j \quad \forall j\in J \tag{4}$$

$$x_{hijk}\ge 0 \quad \forall h\in H, i\in I, j\in J, k\in K \tag{5}$$

$$z_j\in\{0,1\} \quad \forall j\in J \tag{6}$$

Intermediate facility location - MathProgramming

```
set H;
set I;
set J;
set K;

param A{H,I,J};
param B{H,J,K};
param S{H,J};
param F{J};
param Q{H,K};
param W{J};
param Y{H,I};

var x{H,I,J,K}, >= 0;
var z{J}, >= 0, binary;
```

Intermediate facility location - MathProgramming

```
minimize Total_Costs:  sum{h in H, i in I, j in J, k in K}
(A[h,i,j]+B[h,j,k])*x[h,i,j,k] + sum{j in J}F[j]*z[j]
+ sum{h in H, i in I, j in J, k in K}S[h,j]*x[h,i,j,k];

s.t.  R1{h in H, k in K}:  sum{i in I, j in J}x[h,i,j,k] = Q[h,k];
s.t.  R2{h in H, i in I}:  sum{j in J, k in K}x[h,i,j,k] <= Y[h,i];
s.t.  R3{j in J}:  sum{h in H, i in I, k in K}x[h,i,j,k] <=
W[j]*z[j]; solve;
```

MathProgramming - Printing a Report

Good reports bring valuable information

```
printf: "\n========================================\n";
printf: "Intermediate Health Care Units\n";
printf: "========================================\n";
printf: "Logist Cost: $%6.2f\n",
sum{h in H, i in I, j in J, k in K} (A[h,i,j]+B[h,j,k])*x[h,i,j,k];
printf: "Fixed Cost: $%6.2f\n", sum{j in J}F[j]*z[j];
printf: "Variable Cost: $%6.2f\n",
sum{h in H, i in I, j in J, k in K}S[h,j]*x[h,i,j,k];
printf: "========================================\n";
printf: "Total Cost: $%6.2f\n", Total_Costs;
printf: "========================================\n";
printf: "Hospital: Requirement\t Met\n";
printf: "========================================\n";
printf{k in K}: "Hospital [%2d]: %.2f\t%.2f\n", k,
sum{h in H}Q[h,k], sum{h in H, i in I, j in J} x[h,i,j,k];
```

MathProgramming - Printing a Report

```
printf: "========================================\n";
printf: "PHC : Capacity\t Met\tUse(%%)\n";
printf: "========================================\n";
printf{i in I}: "PHC [%2d]: %.2f\t%.2f\t%.2f%%\n", i,
sum{h in H}Y[h,i], sum{h in H, j in J, k in K} x[h,i,j,k],
((sum{h in H, j in J, k in K} x[h,i,j,k])/(sum{h in H}Y[h,i]))*100;
printf: "========================================\n";
printf: "IHC :\tFlow\tUse(%%)\n";
printf: "========================================\n";
printf{j in J: z[j] >= 1}: "IHC [%2d]:\t%.2f\t%.2f%%\n",
j, sum{h in H, i in I, k in K} x[h,i,j,k],
sum{h in H, i in I, k in K} x[h,i,j,k]/(W[j]*z[j])*100;
```

MathProgramming - Printing a Report

```
printf:  "========================================\n";
printf:  "Detailed Flow\n";
printf:  "========================================\n";
printf:  "Pac\tPHC\tIHC\tHosp.\tFlow\n";
printf:  "========================================\n";
printf{h in H, i in I, j in J, k in K: x[h,i,j,k] > 0}:
"[%d]\t[%d]\t[%d]\t[%d]:\t%.2f\n", h,i,j,k,x[h,i,j,k];
printf:  "========================================\n";
```

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